

Original Article

DOI 10.1007/s12206-023-0343-7

Keywords:

- Topology optimization
- Electric vehicles
- Battery thermal management system

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Citation:

Wang, Y., Duong, X. Q., Zadeh, P. G., Chung, J. D. (2023). Liquid-cooled cold plate for a Li-ion battery thermal management system designed by topology optimization. *Journal of Mechanical Science and Technology* 37 (4) (2023) 2079–2086.
<http://doi.org/10.1007/s12206-023-0343-7>

Received September 21st, 2022

Revised December 12th, 2022

Accepted January 2nd, 2023

† Recommended by Editor
Tong Seop Kim

Liquid-cooled cold plate for a Li-ion battery thermal management system designed by topology optimization

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Abstract Modern commercial electric vehicles often have a liquid-based BTMS with excellent heat transfer efficiency and cooling or heating ability. Use of cooling plate has proved to be an effective approach. In the present study, we propose a novel liquid-cold plate employing a topological optimization design based on the globally convergent version of the method of moving asymptotes (GCMMA) method. Comparison with a traditional liquid-cold plate with straight cooling channels revealed that the topology-optimized configuration provides more uniform temperature distribution on the cold plate surface by reducing 48 % of the maximum temperature difference and 57 % of the temperature standard deviation for $Q_0 = 1255 \text{ W/m}^2$. This shows that the topology optimization method is a useful and high-efficiency approach for the innovative design of liquid-cooling plates used for battery thermal management.

1. Introduction

To attain carbon neutrality, an increasing number of countries around the world are promoting the electrification of transportation. High mileage, fast charging, and high performance are all important factors in the development of electric vehicles. Lithium-ion (Li-ion) batteries have gained wide acceptance in the field of electric vehicles due to the advantages of high energy density, long cycle life, high specific power, light weight, low self-discharge rate, and high recyclability compared to other batteries [1].

The performance of Li-ion battery cells is highly sensitive to temperature. Due to internal resistance and chemical interaction, Li-ion batteries create a large amount of heat during charging and discharging, which results in local heat accumulation and temperature rise. This causes capacity fading and thermal runaway [2]. As a result, it is important to keep the battery temperature at a safe level. It was reported that the temperature range should be kept between 25 and 40 °C. Another concern is temperature non-uniformity in the battery module and pack. The temperature difference in a module causes electrical imbalance over time that leads to a state of charge (SOC) mismatch between the cells [3]. Temperature differences inside the cells are required to be less than 5 °C to extend the battery lifespan and provide optimal performance [4]. Therefore, a battery thermal management system (BTMS) is necessary to make sure that the Li-ion batteries operate within the suitable temperature range and to reduce the temperature non-uniformity of the battery and the module.

Liquid cooling BTMS have a larger heat dissipation capacity and commercially developed. Because of its higher cooling capabilities and technical practicality, liquid cooling has proven to be a viable and preferable cooling technology [5]. Moreover, cold plates are widely used between cells in liquid cooling BTMS [6].

For analysis of the thermal performance, a variety of cooling channel designs have been proposed. Huo et al. [7] examined the influence of channel number on the thermal behavior of a cold plate with a straight channel and discovered that increasing the number of channels lowered the maximum battery temperature. Patil et al. [8] investigated the cooling performance

of a lithium-ion battery pouch cell by varying hydraulic diameters with a hexagonal cross section. The results indicated that the effect of hydraulic diameter on the maximum temperature was relatively low. Rao [9] suggested cold plates with a wedge-shaped channel to increase the heat exchange efficiency of a straight channel.

Although extensive studies on the various cooling channel designs have been conducted, a definitive decision based on worldwide standards has rarely been reached. The previous research on BTMS mostly focused on the channel number and the layout of channels [10]. The researchers simply emphasized expanding the diameter of the inlet and outlet sections to reduce the temperature of a cold plate, however this would definitely increase the pressure drops, requiring more power [11].

Topology optimization is normally used in structural mechanics but has been extended to a variety of engineering applications including acoustics, electromagnetics, heat transfer, and fluid flow. This method optimizes the material distribution in a given design domain subject to certain constraints. Dede et al. [12] proposed an ideal design that reduced the domain's thermal resistance and used an additive layer manufacturing approach to fabricate the form for experimental validation. Alexanderson et al. [13] designed passive coolers for light-emitting diode lamps using topology optimization. The improved design lowered the package temperature by 26 % and material use by 12 % compared to a lattice-fin design. Haertel et al. [14] focused on improvement of the condenser air-side surface. First, a simplified 2-D model with fully established internal flow was assumed. The objective function was chosen to maximize heat exchanger conductance, while pressure drop and air temperature change were mandated.

Most of the previous research on battery cold plates focused on their average temperatures. Several studies have explored the effect of uneven temperature distribution inside the battery pack on the performance of Li-ion batteries. Their results showed that temperature non-uniformity led to different charge/discharge characteristics, which resulted in electrical imbalance in a module and degradation of battery pack performance [15]. To overcome the challenges and establish high-efficiency cooling channel construction, a novel flow channel distribution idea based on topology optimization is proposed, with the goal of improving the temperature uniformity distribution of a battery cold plate.

2. Topology optimization

2.1 Governing equations

Heat transfer

$$\rho C_p (\mathbf{u} \cdot \nabla T) = \nabla \cdot (k(\theta) \nabla T) + Q_0 \quad (1)$$

where $k(\theta)$ is the interpolated thermal conductivity as stated in Eq. (6). The term Q_0 is the heat source from battery.

Continuity equation

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

N-S equation

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla P + \eta \nabla^2 \mathbf{u} - \alpha(\theta) \mathbf{u}. \quad (3)$$

The current density model is based on the Darcy inverse permeability idea. Here, θ is a design variable of topology optimization. The essential principle is to add a porous medium resistance term $\mathbf{f} = -\alpha(\theta) \mathbf{u}$ to the N-S equation. This builds the fluid and solid governing equations simultaneously. There is no need to distinguish the different domains during numerical simulation. In Eq. (3), $\alpha(\theta)$ is the material resistance coefficient, which means that the flow resistance is proportional to θ . In this paper, the inverse permeability interpolation method proposed by Borrvall [16] is adopted to control the distribution of the fluid domain and the solid domain by means of controlling local permeability of the medium. The definition is as follows:

$$\alpha(\theta) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot \left(\frac{q \cdot (1 - \theta)}{(q + \theta)} \right), \quad (4)$$

$$\alpha_{\max} = \frac{\eta}{Da \cdot L^2} \quad (5)$$

where $\alpha_{\max}$ is considered a solid material for $\theta = 0$ (i.e., very high flow resistance). The term $\alpha_{\min}$ represents pure fluid material for $\theta = 1$ (i.e., very small flow resistance). Here, $\alpha_{\min}$ is zero, and $\alpha_{\max}$ is defined in Eq. (4). It is determined from the Da number, which is 10^{-5} . Finally, η is the dynamic viscosity of water and q is a tuning parameter for the curvature of $\alpha(\theta)$, which is equal to 0.01.

The same interpolation method was applied to the thermal conductivity.

$$k(\theta) = k_{\min} + (k_{\max} - k_{\min}) \cdot \left(\frac{q \cdot (1 - \theta)}{(q + \theta)} \right) \quad (6)$$

where $k_{\max}$ is the thermal conductivity of aluminum and $k_{\min}$ is the thermal conductivity of water.

2.2 Problem description

In a BTMS, temperature uniformity is the most difficult and important index to achieve. Thus, the standard deviation of temperature T_σ in the design domain was defined as the objective function. The smaller T_σ is, the better the temperature uniformity that could be obtained. Thus, this optimization problem is described as follows:

Find θ
Minimize

$$T_\sigma = \sqrt{\frac{\int (T - T_{\text{avg}})^2 d\Omega_d}{\int d\Omega_d}}. \quad (7)$$

Subject to Eqs. (1)-(3)

$$0 \leq \int \left[\frac{1}{2} \eta \sum_{i,j} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \sum_i \alpha(\theta) u_i^2 \right] \leq 8E_0, \quad (8)$$

$$0 \leq \int \theta d\Omega_d / \int d\Omega_d \leq V_f, \quad (9)$$

$$0 \leq \theta \leq 1.$$

In most situations, the pressure drop of system must be considered in addition to the cooling capacity when designing the cooling system. In the previous work, the power dissipation in Eq. (8) is adopted for the pressure drop [17]. The first term in Eq. (8) is viscous dissipation in a straight channel which is termed as E_0 . The second term is induced by porous medium in the design domain. In a typical topology optimization process, the upper bound of the constraint is eight times that of E_0 . The fluid volume fraction is constrained as the third constraint in Eq. (9).

In the present study, a density model was adopted to perform topological design of 2-D problem. In this density model, a material density variable θ is introduced to control the material distribution. The globally convergent method of moving asymptotes (GCMMA) algorithm proposed by Svanberg [18], which includes adjoint equation and sensitivities analysis, was used to update the design variable distribution. Finally, the design variable was regularized using a Helmholtz filter and the hyperbolic tangent projection, respectively.

2.3 Filter and projection

The density model generally exhibits some numerical problems, such as greyscale elements (meaningless medium design variables) and mesh dependent problems. To avoid such phenomena, regularization was adopted (i.e., Helmholtz filter and hyperbolic tangent projection). In the present study, both methods were applied to penalize the greyscale element to an approximately 0-1 distribution. The PDE based on a Helmholtz filter [19] is expressed as:

$$\theta_f = \theta_c + R_{\min}^{-2} \nabla^2 \theta_f \quad (10)$$

where θ_c is the variable before filtering. The term θ_f is the variable distribution after the filter and $R_{\min}$ is the filter radius which is the minimum mesh size. The Helmholtz filter gives rise to significant grayscale, which does not have a clear physical interpretation. The grayscale can be reduced by applying a hyperbolic tangent projection [20], which is defined as:

$$\theta_h = \frac{(\tanh(\beta(\theta_f - \theta_\beta)) + \tanh(\beta\theta_\beta))}{(\tanh(\beta(1 - \theta_\beta)) + \tanh(\beta\theta_\beta))} \quad (11)$$

where θ_h is the variable after projection and θ_β is the projection point. If it is smaller than θ_β , θ_h will be penalized to zero. Otherwise, the value will approach 1. The term β is the pe-

nalization factor. In this study, θ_β is equal to 0.5 and β is 8.

3. Numerical simulations

Computational fluid dynamics (CFD) was applied to evaluate the thermal-fluidic performance. The commercial CFD software, COMSOL Multiphysics R6.0 was used for topology optimization and numerical simulation. All the CFD simulations work was conducted with Intel Core i9, 2.8 GHz CPU and 32 GB RAM.

3.1 Reference case

In the reference case, the cold plate with six straight channels was inserted between two Li-ion batteries [7]. Due to the significant amount of computing effort and time required, a two-dimensional (2-D) model was assumed. The schematic of the 2-D cold plate and cooling channels is shown in Fig. 1. The outer sizes of L and W are 118 and 63 mm, respectively. The channel width w is 4 mm, and the channels are equally spaced on the cold plate. The depth of a channel h is 1 mm. The Reynolds number was estimated to be 16, and laminar flow was assumed.

The battery temperature varies with the C-rate, which is a measure of the rate at which a battery is being charged or discharged. The rate 1C is known as a one-hour discharge and specifically means that a fully charged battery rated at 1 Ah provides 1 A for one hour [21]. According to Ref. [22], the total heat generation rates of the cell units were different for the different discharge rates. They reported that 1C discharge resulted in maximum heat generation of 40 kW/m³ and that 2C discharge resulted in maximum heat generation of 93 kW/m³. Therefore, six different heat generations were chosen in the range of 40 to 93 kW/m³ and the temperature of the topology optimization model and the reference model were compared.

$$Re_{D_h} = \frac{\rho v_{in} D_h}{\mu}, \quad (12)$$

$$D_h = \frac{4A_c}{P} \quad (13)$$

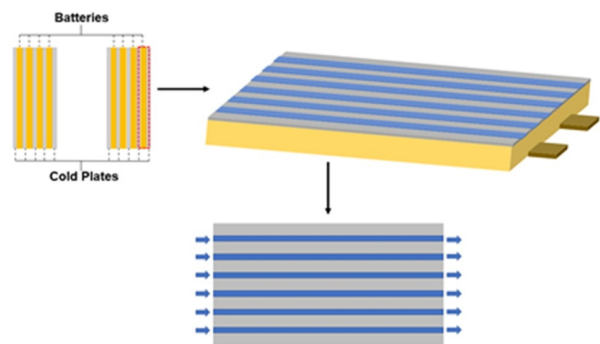


Fig. 1. Schematic of reference straight-channel cold plate.

Table 1. Thermophysical properties of materials.

Material	Properties		
	Density [kg/m ³]	Conductivity [W/m·K]	Dynamic viscosity [Pa·s]
Water	998	0.61	0.001
Aluminum	2700	278	-

where v_{in} is the average inlet velocity, D_h denotes hydraulic diameter, A_c denotes surface area, and P denotes the cross-sectional perimeter. Water was used as the coolant and the cold plate was made of aluminum. The thermos-physical properties of these materials are listed in Table 1.

The mass flow rate of 4×10^{-5} kg/s was used in the reference model [7]. Thus, we chose the inlet velocity of 0.01 m/s, temperature 25 °C, and outlet pressure 0 Pa. The interface had a no-slip wall condition. Heat source Q_0 was imposed on the whole cold-plate domain, in the range from 45 to 91.2 kW/m³.

3.2 Topology optimization case

Symmetric geometry along the horizontal centerline was assumed. The volumetric heat generation of a battery is the same as the reference case. The same boundary conditions in Fig. 1 were applied. Note that the reference uses 6 channels of 4 mm width, but there is one flow inlet of 16 mm width, thus the flow rate is the same for both cases. E_0 in the second constraint is obtained from a straight channel 16 mm wide. The value of viscous dissipation E_0 is 4.42×10^{-6} W/m. The upper bound of power dissipation is $8 E_0$. The maximum iteration number and convergence criteria are 500 and 10^{-4} , respectively. The computation converged because the objective function approaches the minimal value after 188 iterations. The initial value of the design variable is 1; this indicates that the design domain is pure fluid, so the volume fraction V_f is 1 and the power dissipation is very small. With the optimization process, the power dissipation was eventually limited to $8 E_0$, and the volume fraction V_f was limited to the given value (0.38).

The optimization process is shown in Fig. 2, where the dark color represents the solid domain and the light color is the fluid domain. It includes the distribution of design variable θ , the velocity vector, and the temperature at four typical iteration steps. From the initial design, $\theta = 1$ indicates that the design domain is a pure fluid domain. Because the flow path is not still formed, the objective function T_σ is high and the power dissipation is low. From the 1st to 20th iteration, the density model produces the main flow path. Once the main flow path reaches the local optimum, it produces smaller branches until it reaches the optimized design in Fig. 2. Because the objective function is the standard deviation of the temperature, the flow structure will eventually approach the outlet region and achieve the optimized design. Thus, the optimization result will gradually form the structure. After the optimized flow structure forms, the fluid domain could be extracted by defining the design variable to be

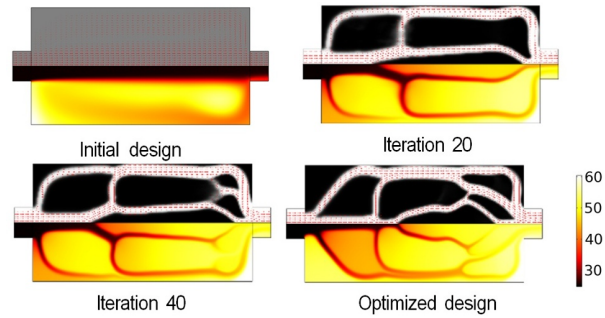


Fig. 2. Optimization process: (upper half) velocity and design variable distribution (black for solid and light for liquid); (lower half) temperature distribution.

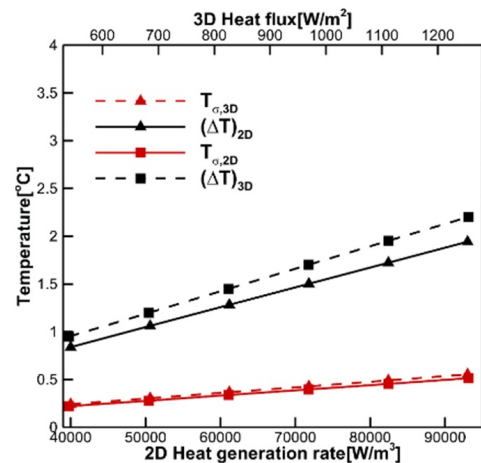


Fig. 3. Comparison of 2D and 3D results.

larger than 0.38.

3.3 Model validation

To confirm the optimization result, we extracted the fluid domain shape obtained from the topology optimization. The temperature standard deviation and temperature difference of extended 3-D topology optimization model are compared with 2-D topology optimization model. The result is shown in Fig. 3. Under the same heat generation from the battery, the temperature standard deviation and temperature difference of 2-D and 3-D results have good agreement with the error of 3 % to 9 %.

A mesh test was conducted for both the reference model (Fig. 1) and the case for topology optimization. The standard deviation of the temperature over the entire domain, for both reference model and topology model, is examined, and the mesh results for each model (31000 and 46330), have error less than 1 %.

4. Results and discussion

4.1 Effect of power dissipation

In the optimization process, the power dissipation of the fluid

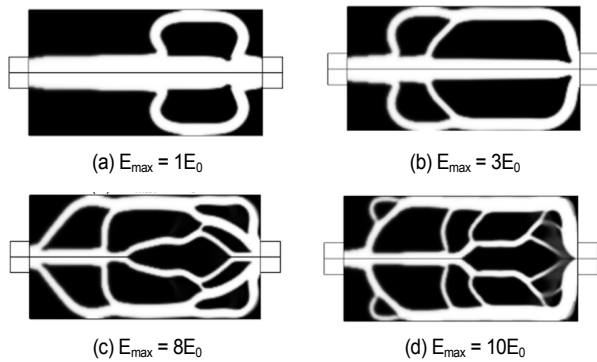


Fig. 4. Optimized design configurations for the different power dissipation.

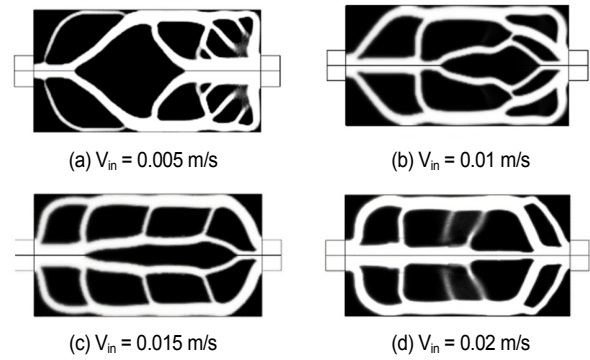


Fig. 6. Optimized design configurations for the different inlet velocity.

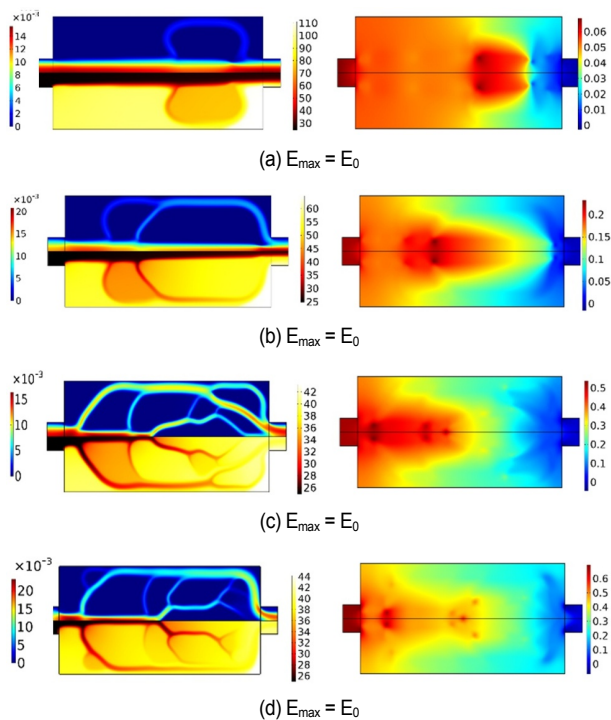


Fig. 5. Velocity profile (upper left); temperature distribution (lower left); pressure drop (right) for the different power dissipation.

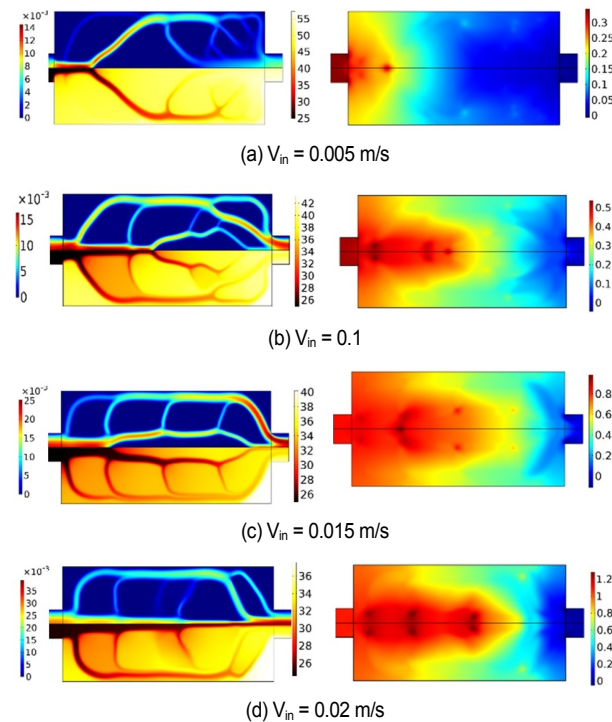


Fig. 7. Velocity profile (upper left); temperature distribution (lower left); pressure drop (right) for the different inlet velocity.

is considered one of the constraints. In Eq. (8), the upper bound of the power dissipation is assumed to be $8E_0$. To evaluate the effect of the power dissipation, four different upper bounds of power dissipation were examined. The power dissipation E_0 in a straight channel was regarded as the reference. The inlet velocity was fixed (0.01 m/s) and the only part changed was the magnitude of the upper bound of the power dissipation from E_0 , $3E_0$, $8E_0$ and $10E_0$. The topological result of $1E_0$ in Fig. 5 shows there only exists three main flow paths after optimization. With increase of the upper bound of power dissipation, more branch paths are produced. The advantage of a branch path is to enhance the heat transfer. However, for large power dissipation, some gray scale will appear, as shown in Fig. 5(d), which does not meet our expectations. The first column of Fig. 5 shows that the maximum temperature

decreases 110 °C, 60 °C, 43.3 °C, and 44 °C, respectively, as the power dissipation increases. On the other hand, the pressure-drop increases to 0.06 Pa, 0.2 Pa, 0.5 Pa and 0.6 Pa, as shown in the second column of Fig. 5. When the power dissipation is set to $8E_0$, the main flow channel and branch paths exist simultaneously and there is no gray scale generated. Therefore, we set the power dissipation to be $8E_0$.

4.2 Effect of inlet velocity

Fig. 6 shows that the optimized structure varies for the different inlet velocities ($v_{in} = 0.005, 0.01, 0.015$ and 0.02 m/s). Here, the upper bound of power dissipation is fixed as $8E_0$. The maximum temperature is reduced as the inlet velocity increases. From first column of Fig. 7, the maximum temperature

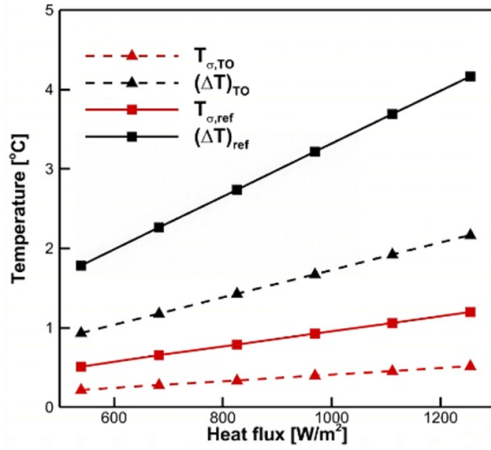


Fig. 8. T_{σ} and $(\Delta T)_{\max}$ comparison with different heat generation.

decreases from 57.3 to 37.6 °C for 0.005 and 0.02 m/s. With the same power dissipation constraint, the topology optimization will not form more branch flow channels.

4.3 Comparison with the reference model

The topology optimization result was compared with the reference model, which has six straight channels at the same inlet velocity of 0.01 m/s. The standard deviation of the temperature over the entire design domain T_{σ} and the maximum temperature difference $(\Delta T)_{\max}$ were chosen to evaluate the thermal performance at different battery discharge rates. The higher discharge rate results in a higher heat generation rate. Thus, as shown in Fig. 8, both T_{σ} and $(\Delta T)_{\max}$ increase with increase of the heat generation rate Q_0 . Due to laminar flow, linear behavior is observed. The difference between the reference model and topology model becomes large at higher Q_0 . When Q_0 is equal to 1255 W/m², the maximum temperature difference $(\Delta T)_{\max}$ of the topology optimization model is decreased by 48 % compared with reference model. The standard deviation of temperature T_{σ} of the topology optimization model is decreased by 57 % compared with reference model. As shown in Fig. 8, the topology optimization greatly improved the temperature uniformity in addition to decreasing the maximum temperature.

5. Conclusions

In the present study, thermal-fluidic topology optimization is conducted to enhance the temperature uniformity on the cold plate surface for electric vehicle batteries. The conclusions are stated as follows.

1) The density-based topology optimization method was used to determine the design variable distribution in the design domain. The standard deviation of temperature on a cold plate was chosen for the objective function. The optimization algorithm GCMMA, the Helmholtz filter, and the hyperbolic tangent projection method were used.

2) The effect of power dissipation and inlet velocity on the optimized configurations were examined. For the different power dissipations (E_0 , $3E_0$, $8E_0$ and $10E_0$), with increase of the power dissipation, more branch paths are produced, the maximum temperature decreased 110, 60, 43.3 and 44 °C, respectively. As the velocity increased from 0.005 to 0.02 m/s, the maximum temperature was 57.3 °C, 43.3 °C, 40.2 °C, and 37.6 °C, respectively. The advantage of a branch path is to enhance the heat transfer. The number of branch channels was highly influential on the temperature uniformity of the cold plate surface.

3) When Q_0 is equal to 1255 W/m², the maximum temperature difference $(\Delta T)_{\max}$ of the topology optimization model is decreased by 48 % compared with reference model. The standard deviation of temperature T_{σ} of the topology optimization model is decreased by 57 % compared with reference model. This indicates that topology optimization is an effective methodology for reducing the maximum temperature and improving the temperature uniformity of a battery cold plate.

Acknowledgments

This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) grant funded by the Korea government (MOTIE) (20202020900060), the Development and Application of Operational Technology in Smart Farm Utilizing Waste Heat from Particulates Reduced Smoke-stack. Yijun Wang is a recipient of China Scholarship Council Scholarship (CSC number: 202208260076).

Nomenclature

A_c	: Cross-sectional area of channel (m ²)
C_p	: Specific heat capacity of fluid (J/kg·K)
D_h	: Hydraulic diameter (mm)
E_0	: Viscous dissipation in straight channel (W/m)
h	: Height of channel (mm)
k	: Thermal conductivity (W/m·K)
$k_{\max}$	: Thermal conductivity of solid (W/m·K)
$k_{\min}$	: Thermal conductivity of liquid (W/m·K)
L	: Length of cold plate (mm)
P	: Cross-sectional perimeter of channel (m)
Q_0	: Volumetric heat source from battery (W/m ³)
q	: Darcy penalization parameter
$R_{\min}$	: Filter radius (mm)
T	: Temperature of design domain (°C)
T_{avg}	: Average temperature of design domain (°C)
T_{σ}	: Temperature standard deviation on cold plate surface (°C)
$\mathbf{u}$	: Fluid velocity field (m/s)
V_f	: Volume fraction of fluid
v_{in}	: Average inlet velocity (m/s)
W	: Width of cold plate (mm)
w	: Width of channel (mm)

Greek symbols

α	: Material resistance coefficient (Pa·s/m ²)
$\alpha_{\max}$	: Solid material resistance coefficient (Pa·s/m ²)
$\alpha_{\min}$	: Fluid material resistance coefficient (Pa·s/m ²)
$(\Delta T)_{\max}$	: Maximum temperature difference on cold plate surface (°C)
η	: Dynamic viscosity of liquid (Pa·s)
θ	: Design variable of topology optimization
θ_p	: Projection point
θ_c	: Variable before filter
θ_f	: Design parameter after PDE filter
θ_h	: Variable after projection
ρ	: Density of fluid (kg/m ³)
Ω	: Design domain

References

- [1] J. Kim, J. Oh and H. Lee, Review on battery thermal management system for electric vehicles, *Applied Thermal Engineering*, 149 (2019) 192-212.
- [2] X. Mo, X. Hu, J. Tang and H. Tian, A comprehensive investigation on thermal management of large-capacity pouch cell using micro heat pipe array, *International Journal of Energy Research*, 43 (13) (2019) 7444-7458.
- [3] F. Bahiraei, A. Fartaj and G. Nazri, Electrochemical-thermal modeling to evaluate active thermal management of a lithium-ion battery module, *Electrochimica Acta*, 254 (2017) 59-71.
- [4] A. Pesaran, Battery thermal models for hybrid vehicle simulations, *Journal of Power Sources*, 110 (2) (2002) 377-382.
- [5] Z. Rao, Z. Qian, Y. Kuang and Y. Li, Thermal performance of liquid cooling based thermal management system for cylindrical lithium-ion battery module with variable contact surface, *Applied Thermal Engineering*, 123 (2017) 1514-1522.
- [6] M. Shen and Q. Gao, Structure design and effect analysis on refrigerant cooling enhancement of battery thermal management system for electric vehicles, *Journal of Energy Storage*, 32 (2020) 101940.
- [7] Y. Huo, Z. Rao, X. Liu and J. Zhao, Investigation of power battery thermal management by using mini-channel cold plate, *Energy Conversion and Management*, 89 (2015) 387-395.
- [8] M. S. Patil, J. H. Seo, S. Panchal, S. W. Jee and M. Y. Lee, Investigation on thermal performance of water-cooled Li-ion pouch cell and pack at high discharge rate with U-turn type microchannel cold plate, *International Journal of Heat and Mass Transfer*, 155 (2020) 119728.
- [9] Z. Rao and X. Zhang, Investigation on thermal management performance of wedge-shaped microchannels for rectangular Li-ion batteries, *International Journal of Energy Research*, 43 (8) (2019) 3876-3890.
- [10] T. Deng, G. Zhang and Y. Ran, Study on thermal management of rectangular Li-ion battery with serpentine-channel cold plate, *International Journal of Heat and Mass Transfer*, 125 (2018) 143-152.
- [11] Y. Huang, P. Mei, Y. Lu, R. Huang, X. Yu, Z. Chen and A. P. Roskilly, A novel approach for lithium-ion battery thermal management with streamline shape mini channel cooling plates, *Applied Thermal Engineering*, 157 (2019) 113623.
- [12] E. M. Dede, S. N. Joshi and F. Zhou, Topology optimization, additive layer manufacturing, and experimental testing of an air-cooled heat sink, *Journal of Mechanical Design*, 137 (11) (2015) 111403.
- [13] J. Alexandersen, O. Sigmund, K. E. Meyer and B. S. Lazarov, Design of passive coolers for light-emitting diode lamps using topology optimization, *International Journal of Heat and Mass Transfer*, 122 (2018) 138-149.
- [14] J. H. Haertel and G. F. Nellis, A fully developed flow thermofluid model for topology optimization of 3D-printed air-cooled heat exchangers, *Applied Thermal Engineering*, 119 (2017) 10-24.
- [15] S. A. Khateeb, S. Amiruddin, M. Farid, J. R. Selman and S. Al-Hallaj, Thermal management of Li-ion battery with phase change material for electric scooters: experimental validation, *Journal of Power Sources*, 142 (1-2) (2005) 345-353.
- [16] T. Borrvall and J. Petersson, Topology optimization of fluids in Stokes flow, *International Journal for Numerical Methods in Fluids*, 41 (1) (2003) 77-107.
- [17] M. Yu, S. Ruan, X. Wang, Z. Li and C. Shen, Topology optimization of thermal-fluid problem using the MMC-based approach, *Structural and Multidisciplinary Optimization*, 60 (1) (2019) 151-165.
- [18] K. Svanberg, *MMA and GCMMA - Two Methods for Nonlinear Optimization*, KTH Royal Institute of Technology (2007) 1-15.
- [19] B. S. Lazarov and O. Sigmund, Filters in topology optimization based on Helmholtz-type differential equations, *International Journal for Numerical Methods in Engineering*, 86 (6) (2011) 765-781.
- [20] F. Wang, B. S. Lazarov and O. Sigmund, On projection methods, convergence and robust formulations in topology optimization, *Structural and Multidisciplinary Optimization*, 43 (6) (2011) 767-784.
- [21] D. Chen, J. Jiang, G. H. Kim, C. Yang and A. Pesaran, Comparison of different cooling methods for lithium ion battery cells, *Applied Thermal Engineering*, 94 (2016) 846-854.
- [22] M. Xu, Z. Zhang, X. Wang, L. Jia and L. Yang, A pseudo three-dimensional electrochemical-thermal model of a prismatic LiFePO₄ battery during discharge process, *Energy*, 80 (2015) 303-317.



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